

Vulnerability Assessment Report

May 7th 2026

System Description

The server hardware consists of a powerful CPU processor and 128GB of memory. It runs on the latest version of Linux operating system and hosts a MySQL database management system. It is configured with a stable network connection using IPv4 addresses and interacts with other servers on the network. Security measures include SSL/TLS encrypted connections.

Scope

The scope of this vulnerability assessment relates to the current access controls of the system. The assessment will cover a period of three months, from June 2026 to August 2026. [NIST SP 800-30 Rev. 1](#) is used to guide the risk analysis of the information system.

Purpose

The database server is a centralized computer system that stores and manages large amounts of confidential and sensitive data. The server is used to store customer, campaign, and analytic data that can later be analyzed to track performance and personalize marketing efforts. It is critical to secure the system because of its regular use for marketing operations.

Risk Assessment

Threat source	Threat event	Likelihood	Severity	Risk
Hacker	Obtain sensitive information via exfiltration	3	3	9
Employee	Alter/Delete critical information	2	3	6
Customer	Perform reconnaissance and surveillance of organization	1	3	3

Approach

Risks considered the data storage and management methods of the business. The likelihood of a threat occurrence and the impact of these potential events were weighed against the risks to day-to-day operational needs.

All three threats involve the tampering or stealing of private protected data of customers, which falls under the strict regulations of mandatory policies. With the server being open to the public, one of these threats can be written off. If this data is tampered with or breached, it will cause the business to take a huge blow in both their reputation with customers and incur large fines to pay for violating the mandatory regulations. These factors can severely impact the business's continuity.

Remediation Strategy

Implementation of authentication, authorization, and auditing mechanisms to ensure that only authorized users access the database server. This includes using a password policy to ensure strong passwords, role-based access controls to limit user privileges, and multi-factor authentication. Encryption of data in motion using TLS. IP allow-listing to corporate offices to prevent random users from connecting to the database.